CAMBRIDGE INFOTECH

API and JSONs

Assignment

Understanding APIs for Corporate Employees

NAME Click here to type your name	EMPLOYEE ID Click here to type ID	DATE Click here to type date
TOTAL QUESTIONS 30	DURATION 60 Minutes	TOTAL MARKS 70

Instructions

- This assignment contains 30 questions across five sections: Multiple Choice, True/False, Fill in the Blanks, Short Answer, and Practical/Coding.
- Type your answers directly into this document — in the blank fields, answer lines, or code boxes provided under each question.
- You may refer to official documentation, but all answers must be written in your own words / own code.
- Total marks: 70. Suggested duration: 60 minutes.
- Where an API endpoint is needed, use https://jsonplaceholder.typicode.com unless another URL is specified.

Section A — Multiple Choice Questions 10 marks

Choose the single best answer. (1 mark each)

1. What does API stand for?

A) Application Programming Interface
B) Automated Program Integration
C) Application Process Interaction
D) Advanced Programming Instruction

Your Answer: ___A_______

2. In the client–server model, which of these is typically the "client"?

A) A database
B) A web browser or mobile app
C) A web server
D) An operating system kernel

Your Answer: __B________

3. Which HTTP method is used to retrieve data without modifying it?

A) POST
B) GET
C) DELETE
D) PUT

Your Answer: ___B_______

4. What does an HTTP status code of 404 mean?

A) Internal server error
B) Unauthorized access
C) Resource not found
D) Request succeeded

Your Answer: ____C______

5. Which HTTP method replaces an entire existing resource?

A) PATCH
B) PUT
C) GET
D) POST

Your Answer: ____B______

6. What data format do most modern web APIs use to exchange data?

A) XML only
B) CSV
C) JSON
D) Plain text

Your Answer: _____C_____

7. Which status code indicates a resource was successfully created?

A) 200
B) 201
C) 301
D) 404

Your Answer: _____B_____

8. In Python, which library is most commonly used to send HTTP requests?

A) pandas
B) requests
C) numpy
D) flask

Your Answer: _____B_____

9. What does response.json() do?

A) Sends JSON data to the server
B) Converts the response body into a Python dictionary
C) Permanently deletes JSON data
D) Encrypts the JSON payload

Your Answer: ____B______

10. Which HTTP header is typically used to send authentication credentials?

A) Content-Type
B) Authorization
C) Accept
D) User-Agent

Your Answer: ___B_______

Section B — True or False 5 marks

Type True or False. (1 mark each)

11. A GET request can safely be sent multiple times without changing data on the server.

Your Answer (True / False): ____True________

12. JSON stands for JavaScript Object Notation.

Your Answer (True / False): _____True_______

13. A Python dictionary and a JSON object are exactly the same thing in every way.

Your Answer (True / False): _____False_______

14. PATCH updates an entire resource, exactly like PUT.

Your Answer (True / False): ___False________

15. A requests.Session() allows multiple requests to reuse the same underlying connection.

Your Answer (True / False): ___True_________

Section C — Fill in the Blanks

5 marks

Type the missing word or term. (1 mark each)

16. The HTTP status code _______________ means "Unauthorized."

Your Answer: ____401____________

17. The _______________ HTTP method is used to permanently remove a resource from a server.

Your Answer: ______DELETE__________

18. In a URL, the part after the domain that identifies a specific resource is called the _______________.

Your Answer: _____Path___________

19. The Python function requests._______________() is used to send a GET request.

Your Answer: ____get____________

20. A _______________ parameter is used to filter or customize a GET request without changing the endpoint itself.

Your Answer: ____Query____________

Section D — Short Answer Questions 15 marks

Answer in 2–4 sentences. (3 marks each)

21. Explain the difference between PUT and PATCH, and give one example of when you would use each.

PUT updates entire existing data with new data, while PATCH Updates only specific fields of resource.

Example:

PUT: Updating complete user profile with name, user name, email, and address.

PATCH: Updating only the user Email ID while keeping other details unchanged.

22. Describe the request–response cycle in your own words, using a real-life analogy (for example, a restaurant or a pizza order).

Request- Response cycle is like ordering food at a restaurant. A customer(client) gives an order(request) to the waiter, who sends it to the kitchen(Server). The server prepares food and sends it back as the response. Similarly a client sent an API request and server responds back for the request.

23. Why is it unsafe to hardcode an API key directly in your code? What is the safer alternative?

Hardcoding API key directly in the code is unsafe because anyone accessing the code may steal or misuse the key. The safer approach is storing API keys in Environment variables or secure configuration files.

24. What is the purpose of setting a timeout on an API call? What happens if you don't set one?

A Timeout sets the maximum waiting time for a server response. Without a timeout, the program may keep waiting forever if the server is slow or unavailable, causing the application to freeze.

25. Explain what a Session object does in the requests library and why it can improve performance.

A session object stores setting like headers and cookies across multiple requests. It improves performance by reusing the same connection instead of creating the new connection every time.

Section E — Practical / Coding Questions 20 marks

Type working Python code into the box below each question.

26. Write Python code using the requests library to send a GET request to https://jsonplaceholder.typicode.com/posts and print the response status code.

```
import requests
url = "https://jsonplaceholder.typicode.com/posts"
response = requests.get(url)
print(response.status_code)
```

27. Write Python code to send a POST request to https://jsonplaceholder.typicode.com/posts with a JSON body containing a "title" and a "body" field, then print the JSON response.

Hint: Use json=your_dictionary as the second argument to requests.post().

```
import requests
url = "https://jsonplaceholder.typicode.com/posts"
new_post = {
    "title": "Learning API",
    "body": "The content has been updated."
}
response = requests.post(url, json=new_post)
print(response.json())
```

28. Write Python code that uses query parameters to fetch all posts where userId equals 2.

Hint: Build a params dictionary and pass it with params=params.

```
import requests
url = "https://jsonplaceholder.typicode.com/posts"
params = {
  "userId": 2
}
response = requests.get(url, params=params)
print(response.json())
```

29. Write a try/except block that sends a GET request with a 5-second timeout, and prints "Request Timed Out" if the server does not respond in time.

Hint: Catch requests.exceptions.Timeout.

```
import requests
try:
    url = "https://jsonplaceholder.typicode.com/posts"
    response = requests.get(url, timeout=5)
    print(response.json())
except requests.exceptions.Timeout:
    print("Request Timed Out")
```

30. Convert the JSON response from https://jsonplaceholder.typicode.com/posts into a pandas DataFrame and display the first 5 rows.

Hint: Use pd.DataFrame(response.json()) and .head().

```
import requests
import pandas as pd
url = "https://jsonplaceholder.typicode.com/posts"
response = requests.get(url)
data = response.json()
df = pd.DataFrame(data)
print(df.head())
```

— End of Assignment —